INCH-POUND

MIL-C-27072/94B
23 March 1993
SUPERSEDING
MIL-C-27072/94A
16 June 1989

MILITARY SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL,
SINGLE CONDUCTOR, SINGLE SHIELDED
M27072/94*DDE*

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and and Standards (DODISS) specified in the solicitation: MIL-C-27072.

Inactive for new design. For new design use MIL-C-55021/2.

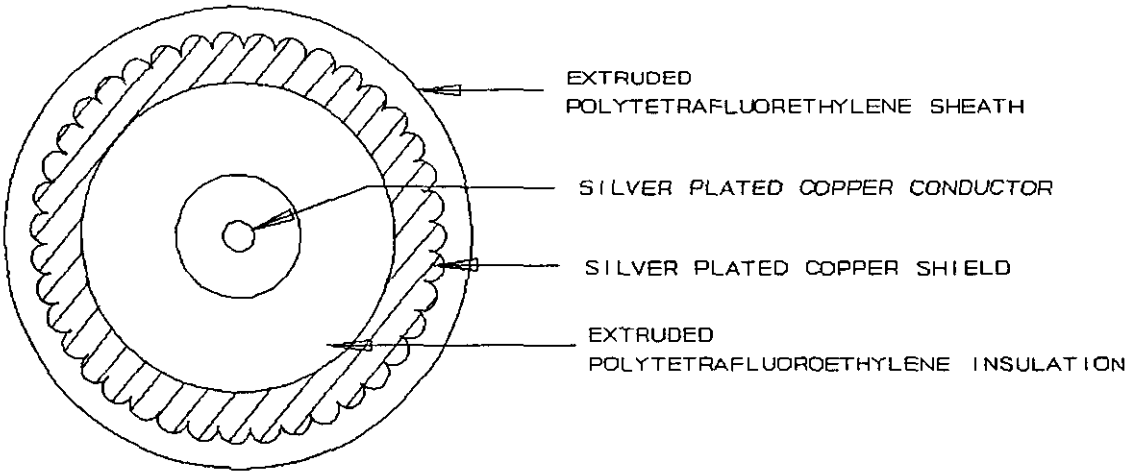


FIGURE 1. Cable configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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DESIGN DATA:

Design and construction: See figure 1.

Number of conductors: One.

Conductor size and stranding: See table I.

Wire: In accordance with M16878/4, or /21 (see table I for complete Part or Identifying Number (PIN)).

Jacket undershield: Not applicable.

Shielding: Silver coated copper

Sheath: Polytetrafluoroethylene.

REQUIREMENTS:

Visual and dimensional:

Components:

Basic wire: Silver coated copper

Dimension: See table I.

Primary insulation:

Material: Polytetrafluoroethylene.

Color: White.

Jacket over insulation: Not applicable.

Wall thickness: Not applicable.

Color: Not applicable.

Spark test voltage: 3,400 V rms.

Inspection test voltage: 2,000 V rms.

Impulse dielectric test: 6,500 V peak.

Filler: Not applicable.

Material: Not applicable.

Lay: Not applicable.

Shield:

Material: Silver coated copper.

Gauge: 38 AWG.

Braid angle: 40°C maximum, 20°C minimum.

Workmanship: In accordance with MIL-C-27072.

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Sheath:

Material: Polytetrafluoroethylene.

Tensile strength: In accordance with table V of MIL-C-27072.

Elongation: In accordance with table I of MIL-C-27072.

Tension set: Not applicable.

Tear strength: Not applicable.

Accelerated aging: Not applicable.

Oil resistance: Not applicable.

Finished cable:

Dimension: In accordance with table I.

Conductor resistance: In accordance with MIL-W-16878/4 or /21.

Dielectric strength: In accordance with MIL-W-16878/4 or /21.

Cold bend: Mandrel size and speed in accordance with table XIII of MIL-C-27072.

Ozone: Mandrel size in accordance with table XIV of MIL-C-27072.

Marking durability: In accordance with MIL-C-27072.

Heat Stability: Not applicable.

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TABLE I. Construction data.

Pin M27072/94	Wire size	Number of Strands	Strand size (AWG)	Nominal diameter (inches)	Finished cable diameter (inches)		Basic wire M16878/4 M16878/21
					Min	Max	
EDDE*	14	19	27	.072	.125	.152	BKE9
FDDE*	16	19	29	.057	.111	.138	BJE9
GDDE*	18	19	30	.050	.102	.121	BHE9
JDDE*	20	19**	32	.041	.095	.112	BGE9
KDDE*	20	7	28	.038	.092	.109	BGB9
LDDE*	22	19**	34	.032	.086	.103	BFE9
MDDE*	22	7	30	.030	.084	.101	BFB9
NDDE*	24	19**	36	.035	.076	.098	BEE9
PDDE*	24	7	32	.024	.073	.095	BEB9
SDDE*	26	7	34	.019	.068	.090	BDB9
TDDE*	28	7	36	.015	.064	.086	BCB9
WDDE*	30	7	38	.012	.061	.083	BBB9

* Asterisk (*) shall be replaced with the appropriate color of the cable sheath in accordance with MIL-STD-681 Shall be replaced with the appropriate color of the cable sheath in accordance with MIL-STD-681, table I. If a strip or a band is to be added it shall be in accordance with 3.2.2 of MIL-C-27072.

** Preferred stranding number for wire size.

NOTE: These constructions supersede those shielded single conductor cables of the same construction formerly covered by MIL-W-16878.

Revision changes are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Air Force - 85
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6145-2025-03)

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